如何和拥有神话武器的小孩一起玩玩具

SPS 单案例：Excel 直出结果

Moxian Qian
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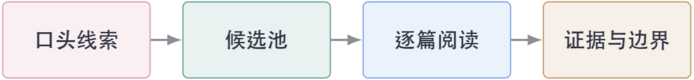
GitHub: github.com/qxxmax/skillforpaper

目录

调用

调用句	请使用 play-the-toy-with-children 做 Part 1 文献调研。
口头线索	Moxian Qian 最近的 stochastic path sampler 文章。
核心要求	确认对象；多轮扩张；逐篇阅读；回到原文；保留来源、边界和待确认问题。
输出	source table；reading notes；claim/boundary ledger；lineage；gaps；report。



对象确认

字段	已确认结果
题名	Stochastic Path Sampler For Lattice Field Theory
作者	Shiyang Chen; Moxian Qian; Gert Aarts; Biagio Lucini; Kai Zhou
arXiv	2606.13790v1
日期	2026-06-11
主来源	arXiv 页面和下载 PDF
顺序	先锁定身份，再扩张引用与方法谱系。

层级	进入条件	能做什么
C0	搜索召回	只算候选。
C1	题名/作者/年份/ID	进入候选池。
C2	页面或 PDF 可打开	确认对象存在。
C3	原文上下文或方法位置	进入谱系判断。
C4	claim + 页码/图表/公式	才能写入报告。

搜索扩张

	A	B	C	D	E	F
1	query_id	family	source	query	facet	parent_evidence
2	Q01	root	OpenAlex	"Stochastic Path Sampler For Lattice Field Theory"	identity	root title
3	Q02	root	Crossref	"Stochastic Path Sampler For Lattice Field Theory"	identity	root title
4	Q03	root	arXiv	ti:"Stochastic Path Sampler For Lattice Field Theory"	identity	root title
5	Q04	backward	root_bibliography	all live root references	lineage	root TeX
6	Q05	forward	OpenAlex	works citing arXiv:2606.13790	forward	root arXiv ID
7	Q06	author	OpenAlex	"Shiyang Chen" lattice sampler	author	root title page
8	Q07	author	OpenAlex	"Moxian Qian" lattice sampler	author	root title page
9	Q08	author	OpenAlex	"Gert Aarts" lattice sampler	author	root title page
10	Q09	author	OpenAlex	"Biagio Lucini" lattice sampler	author	root title page

- 每条 query 都留下 family、source、facet 和 parent evidence。

候选筛选

candidate_id	title
arxiv:2606.13790	Stochastic Path Sampler For Lattice Field Theory
arxiv:2502.05504	Physics-Conditioned Diffusion Models for Lattice Gauge Theory
arxiv:2409.15937	Numerical determination of the width and shape of the effective string using Stochastic Normalizing Flows
arxiv:2412.19109	Stochastic normalizing flows for Effective String Theory
arxiv:2210.03139	Stochastic normalizing flows for lattice field theory

score	decision
121	include
20	include
19	include
18	include
18	include

最终候选池有 593 条；screening 决定先读谁，但候选本身还不是证据。

Dijkstra 排序

	A	B	C	D	E
1	metric	relevance_only	candidate_dijkstra	final_with_gap_closure	interpretation
2	selected papers	30	30	37	fixed reading budget for the first two arms
3	declared facets covered	7	7	10	Dijkstra did not increase coarse facet count
4	method groups covered	8	8	8	coverage equality does not imply identical papers
5	exact root-bibliography papers	19	21	21	candidate Dijkstra retained two more root ancestors
6	mean heuristic screen score	16.7	16.067	not comparable	graph navigation traded a little keyword score for paths
7	selection overlap	20	20	35	20/30 papers overlap; ten are replaced

同样读 30 篇时，Dijkstra 多保留 **2 篇目标论文的直接参考文献**； coarse facets 和 method groups 没有增加。

来源表

paper_id	arxiv_id	title
P001	2606.1379	Stochastic Path Sampler For Lattice Field Theory
P002	2111.15141	Path Integral Sampler: a stochastic control approach for sampling
P003	2302.13834	Denoising Diffusion Samplers
P004	2307.0105	Transport meets Variational Inference: Controlled Monte Carlo Diffusions
P005	2410.02711	NETS: A Non-Equilibrium Transport Sampler
P006	2210.03139	Stochastic normalizing flows for lattice field theory
P007	2201.08862	Stochastic normalizing flows as non-equilibrium transformations

source_relation	source_url
root	https://arxiv.org/abs/2606.13790
direct_bibliography	https://arxiv.org/abs/2111.15141
direct_bibliography	https://arxiv.org/abs/2302.13834
direct_bibliography	https://arxiv.org/abs/2307.01050
direct_bibliography	https://arxiv.org/abs/2410.02711
direct_bibliography	https://arxiv.org/abs/2210.03139
direct_bibliography	https://arxiv.org/abs/2201.08862

最终选择并逐篇阅读 37 篇全文；每行同时保留身份、关系和 primary URL。

问题与方法

	I	J
1	problem	problem_anchor
2	Data-free sampling of an unnormalized lattice target near criticality, where ordinary chains develop long autocorrelation times.	PDF p.1 abstract
3	Construct an efficient sampler for an unnormalized target by controlling a stochastic path rather than fitting a terminal density alone.	PDF p.1 abstract

	K	L
1	method	method_anchor
2	Learn paired forward and auxiliary backward Langevin paths by minimizing a path-space variational free energy / entropy-production upper bound.	PDF pp.4-8 Sec. 2
3	Formulate sampling as a Schrodinger-bridge / stochastic optimal-control problem and learn a neural control for a reference diffusion.	PDF pp.4-8 method

problem 和 method 都必须带 PDF 页码或 section anchor。

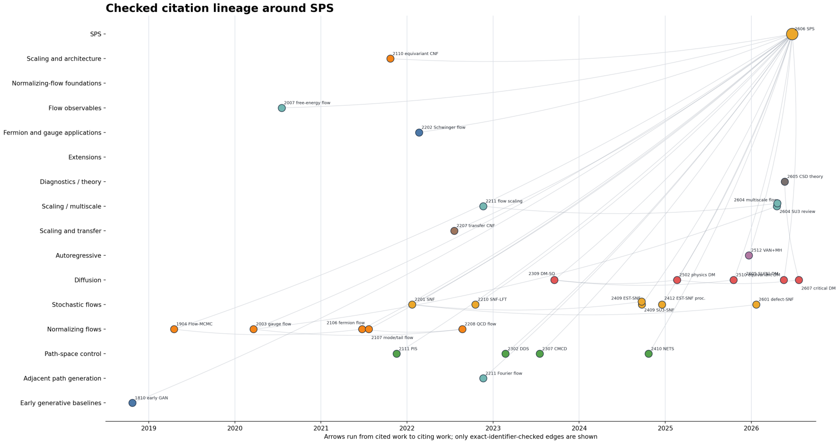
结果与局限

	M	N	O
1	correction_exactness	result	result_anchor
2	Independent path proposals are corrected with an extended-space independence Metropolis-Hastings step;	Corrected SPS observables agree with HMC in the tested 2D phi4 systems and show shorter autocorrelation near the pseudocritical region.	PDF pp.12-18 Sec. 3, Tables 1-4
3	Girsanov path likelihood ratios provide importance weights; weighted estimates retain the path-measure correction.	The learned path sampler is competitive with the reported MCMC, SMC and normalizing-flow baselines on its benchmark suite.	PDF pp.9-18 experiments

	P	Q
1	limitation	limitation_anchor
2	Uncorrected tails and multimodal regions deviate; finite network capacity, terminal diffusion and support coverage remain practical limits, and the reported timing is hardware/setup specific.	PDF pp.12-13 Secs. 3.1-3.2; pp.19-21 discussion
3	Training overhead, time discretization and a suboptimal learned control can produce weight variance and residual bias if correction is omitted.	PDF pp.19-20 discussion

exactness、result、limitation 和各自原文位置保存在同一行。

引用谱系



横轴是年份，纵轴是方法家族；箭头从被引工作指向引用它的工作。

结论与边界

	A	B	C	D	E	F
1	claim_id	claim	allowed_short_sentence	evidence_ids	boundary	status
2	C01	SPS learns paired forward and auxiliary backward stochastic paths with a path-space variational objective.	SPS learns a low-irreversibility path between prior and target.	E-260613790-M	This is the SPS v1 formulation; do not replace it with terminal-density KL training.	confirmed
3	C02	SPS follows learned independent path proposals with extended-space independence Metropolis-Hastings.	The learned SPS proposal is followed by an exactness gate.	E-260613790-M;E-260613790-R	Do not call uncorrected SPS samples exact.	confirmed
4	C03	The nearest method lineage combines path-space stochastic control, diffusion samplers and nonequilibrium transport.	SPS sits at the intersection of path-space control and nonequilibrium transport.	E-211115141-M;E-230213834-M;E-230701050-M;E-241002711-M;E-260613790-M	This is a checked synthesis, not a claim that the algorithms are identical.	confirmed
5	C04	The lattice lineage repeatedly pairs learned proposals with Metropolis, reweighting or Jarzynski correction.	Learned lattice proposals are repeatedly paired with an explicit correction mechanism.	E-190412072-M;E-200306413-M;E-210605934-M;E-220108862-M;E-220803832-M	Shared correction logic does not imply equal scaling or performance.	confirmed
6	C05	Diffusion models do not automatically remove critical slowing; architecture and locality control the outcome in the checked theory study.	Diffusion helps only when architecture and physics match the slow modes.	E-260512597-R;E-260512597-L;E-260708505-L	The quadratic-to-logarithmic result is controlled in the Gaussian O(n to infinity) setting, not universally.	confirmed
7	C06	Cross-volume scores, coarse-to-fine factorization and localized defect maps are distinct scale-transfer routes.	Transfer, multiscale structure and locality are three distinct routes to larger lattices.	E-260708505-R;E-260410209-R;E-260120708-R	Each result is benchmark-specific; production-QCD scaling is not established.	confirmed
8	C07	A fair efficiency claim must include training, generation, correction and observable-level uncertainty.	Efficiency means physical error per total compute, not a single ESS number.	E-221107541-R;E-221107541-L;E-190412072-L;E-260410209-R;E-240918861-L	The literature does not yet supply one uniform cross-paper cost benchmark.	confirmed
9	C08	Aggregate loss or ESS can hide support, tail, multimodal and infrared errors.	A sampler can look good globally and still miss the modes that matter.	E-210700734-R;E-210700734-L;E-260708505-R;E-260613790-L	Diagnostics expose risk; they do not by themselves correct it.	confirmed
10	C09	Realistic lattice-QCD evidence remains sparse despite several gauge, fermion and 4D SU(3) demonstrations.	The field is active, but production-scale lattice-QCD evidence is still sparse.	E-260412416-R;E-260412416-L;E-220803832-L;E-260120708-R	Do not describe these methods as mature at production QCD scale.	confirmed
11	C10	The defensible SPS follow-ups close correctness, scaling and physics-cost loops together.	The next SPS step is to close correctness, scaling and physics-cost loops together.	E-260613790-L;E-210700734-L;E-260512597-L;E-260708505-L;E-260410209-L;E-210605934-L	These are literature-grounded directions, not SPS results already achieved.	confirmed

claim、可写短句、evidence ID、boundary 和 status 位于同一张表。

Backup：运行记录

记录项	最终实跑结果
candidate graph	3,300 nodes / 8,187 edges；本地 Dijkstra 约 0.11 s。
equal reading budget	relevance-only 30 篇；Dijkstra 30 篇；重合 20 篇。
selection effect	exact root-bibliography papers：19 $\rightarrow$ 21。
gap closure 后	37 篇全文；10 个 facets；185 条 evidence；199 条已检查直接引用关系。
complete goal	1,012,083 tokens；4,039 s；19/19 validation gates PASS。
不能推出	没有分别测量两个 ranking arm 的 token/time；Dijkstra path 是导航元数据，不是科学证据。

算法决定先查哪里；原文和 evidence gate 决定最后能写什么。